reproducing property

$$f(x) = \sum_{i=1}^{n} \alpha_i k(x, x_i)$$

$$= \sum_{i=1}^{n} \alpha_i \langle k(\cdot, x_i), k(\cdot, x) \rangle$$

$$= \left\langle \sum_{i=1}^{n} \alpha_i k(\cdot, x_i), k(\cdot, x) \right\rangle$$

$$= \langle f(\cdot), k(\cdot, x) \rangle$$

$\beta, \gamma \in X$

$k(x, y) = x^T y$

$f(x) = x^T \beta = \langle x, \beta \rangle$

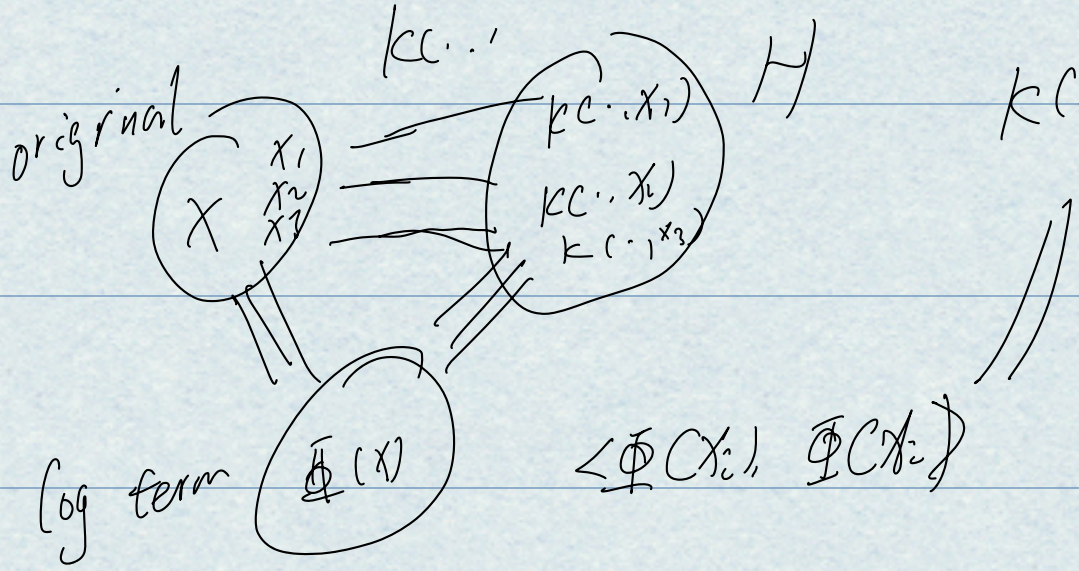
$= \langle x, \sum e_i \beta_i \rangle$

$$= \sum_{i=1}^{p} \beta_i \langle x, e_i \rangle$$

$$= \sum_{i=1}^{p} \beta_i x_i$$

$$= x^T \beta$$

kernel ridge regression



$$|f(x) - f(x')|$$

$$= |\langle f, k(\cdot, x) \rangle - \langle f, k(\cdot, x') \rangle|$$

$$= |\langle f, k(\cdot, x) - k(\cdot, x') \rangle_H|$$

gausim kernal

$$\exp\left(-\frac{\|x - x_1\|^2}{\sigma^2}\right)$$

$$\leq \|f\|_H \cdot \|k(\cdot, x) - k(\cdot, x')\|_H$$

$$\sqrt{\langle k(\cdot,x)-k(\cdot,x'),\, k(\cdot,x)-k(\cdot,x')\rangle}$$

$$= \sqrt{k(x,x)+k(x',x')-2k(x,x')}$$

$$\leq \sqrt{2}$$

fmd kernal

RKHS $\Rightarrow$ $\frac{K}{\uparrow}$.

Ridge Reg.

$$\arg \frac{1}{2n\lambda}\|y-X\beta\|_2^2+\frac{1}{2}\|\beta\|_2^2$$

subj $z_i = y_i - x_i^T\beta$

$$\min_{\theta} g(\theta)$$

subj $h_i(\theta)=0 \quad \forall i$

$$L(\theta, \alpha) = g(\theta) + \sum_{i=1}^{n} \alpha_i h(\theta)$$

$$\min_{\theta} \max_{\alpha} L(\theta, \alpha)$$

可以不用 choose θ, 而选 α 会无限.

$\theta \in (\beta, z)$

$$L() = \frac{1}{2n\lambda} \|z\|_2^2 + \frac{1}{2} \|\beta\|_2^2 + \sum_{i=1}^{n} \alpha_i (y_i - x_i^T \beta - z_i)$$

$$\frac{\partial L}{\partial z} = \frac{1}{n\lambda} z_i - \alpha_i \overset{\text{set}}{=} 0 \Rightarrow z_i = n\lambda \cdot \alpha_i$$

$$\frac{\partial L}{\partial \beta} = \beta - \sum_{i=1}^{n} \alpha_i \cdot x_i \overset{\text{set}}{=} 0 \Rightarrow \beta = \sum_{i=1}^{n} \alpha_i x_i$$

plug back into $L(\theta, \alpha)$

$$\frac{n\lambda}{2} \alpha^T \alpha + \frac{1}{2} \sum_{i=1}^{n} \sum_{j=1}^{n} \alpha_i \alpha_j x_i^T x_j$$

$$+ \sum_{i=1}^{n} \alpha_i y_i - \sum_{i=1}^{n} \alpha_i x_i^T \left(\sum_{j=1}^{n} \alpha_j x_j \right) - n\lambda \cdot \alpha^T \alpha$$

$$= -\frac{n\lambda}{2}\alpha^T\alpha - \frac{1}{2}\sum_{i,j}\alpha_i\alpha_j \cdot x_i^T x_j + \alpha^T y.$$

$$\sum_{i,j}\alpha_i\alpha_j\, k(x_i, x_j)$$

$$= \alpha^T K_{n\times m}\alpha$$

$$L(\alpha) = \frac{n\lambda}{2}\alpha^T\alpha - \frac{1}{2}\alpha^T K\alpha + \alpha^T y.$$

$$\frac{\partial L}{\partial \alpha} = -n\lambda\alpha - K\alpha + y \overset{\text{set}}{=} 0$$

$$y = n\lambda\alpha + K\alpha$$

$$= (n\lambda I + K)\alpha$$

$$\boxed{\alpha = (n\lambda I + K)^{-1} y}$$

linear obvious / general is this too.

$$f(x) = \beta^T x$$

$$= \sum_{i=1}^{n} \alpha_i x_i^T x$$

$$= \sum_{i=1}^{n} \alpha_i k(x_i, x)$$

$$f(\cdot) = \sum_{i=1}^{n} \alpha_i k(x_i, \cdot).$$

RKHS. 不在乎 Care about Linear.

$$\min_{f \in H} = \frac{1}{n} \sum (y_i - f(x_i))^2 + \lambda \|f\|_H^2$$

$$f = \sum_{i=1}^{n} \alpha_i k(\cdot, x)$$

$$\alpha = (n\lambda) I$$